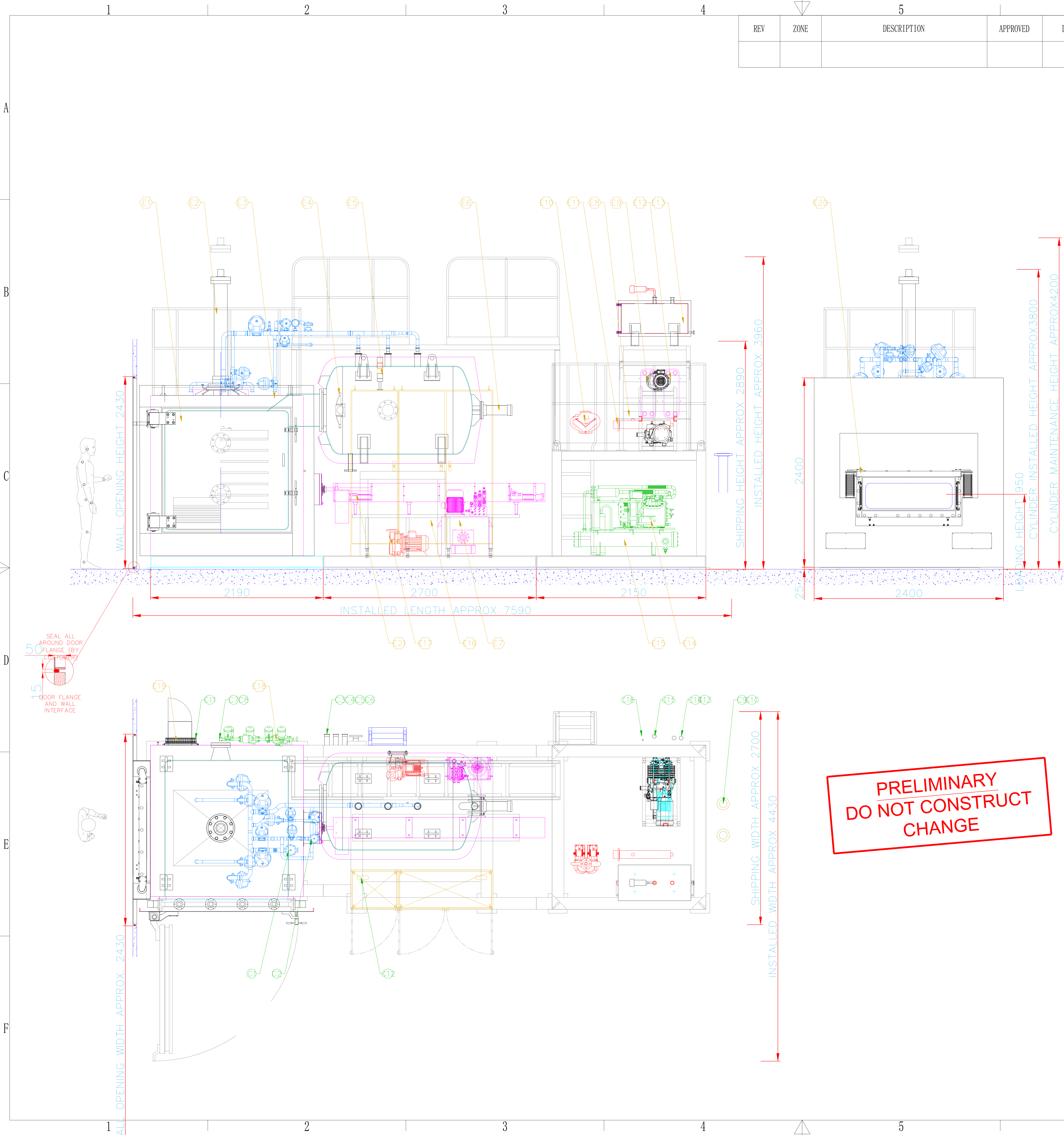




REV	ZONE	DESCRIPTION	APPROVED	DATE

UTILITY REQUIREMENTS (APPROX.)

UTILITY #	DESCRIPTION	QUANTITY	REMARK
1	ELECTRICITY-TOTAL ELECTRICAL	58KW	380V/50 HZ,3 PHASE,5 WIRE
2	COOLING WATER FOR REFRIGERANT COOLING	≥16M ³ /HR	1.5≤P≤3BAR,15℃≤T≤30℃
3	CITY WATER FOR JACKET COOLING	≥40L/MIN	1.5≤P≤2BAR,12℃≤T≤25℃
4	CITY WATER FOR WATER RING PUMP	≥15L/MIN	1.5≤P≤2BAR,12℃≤T≤25℃
5	WFI FOR CLEANING	≥220L/MIN	3≤P≤5BAR,T≥80℃
6	COMPRESSED AIR FOR PNEUMATIC VALVE	≥40L/MIN	5≤P≤8BAR
7	PURE STEAM CONSUMPTION FOR SIP	≥200KG/BATCH	2≤P≤3BAR
8	PURE STEAM CONSUMPTION FOR DE-ICING	≥40KG/BATCH	2≤P≤3BAR
9	CLEAN AIR FOR AERATION SYSTEM	≥40L/MIN	0.5≤P≤1BAR
10	CLEAN N ₂ FOR AERATION SYSTEM	≥40L/MIN	0.5≤P≤1BAR
11	COOLING WATER FOR VACUUM SYSTEM	≥6L/MIN	2≤P≤3BAR,15℃≤T≤30℃
12	DRY N ₂ FOR VACUUM SYSTEM	≥40L/MIN	P=4.5BAR

TECHNICAL SPECIFICATIONS SUMMARY

SR NO.	DESCRIPTION	QUANTITY	REMARK
1	TOTAL NET SHELF AREA	5.43M ²	
2	SHELF GROSS DIMENSIONS(D)	1200MM	
3	SHELF GROSS DIMENSIONS(W)	960MM	
4	QUANTITY OF SHELVES	5+1	
5	SHELVES SPACING	120MM	
6	SHELVES TEMPERATURE RANGE	-55 TO +80℃	
7	SHELF COOLING RATE	+20 TO -40℃≤60MIN	EMPTY LOAD
8	SHELF HEATING RATE	1℃/MIN	EMPTY LOAD
9	CONDENSER COIL COOLING RATE	+20 TO -40℃≤30MIN	EMPTY LOAD
10	MINIMUM COILED TEMPERATURE	-75℃	
11	MAXIMUM ICE CAPACITY	100KG/BATCH	
12	ULTIMATE VACUUM	0.01MBAR	
13	EVACUATION RATE	FROM 1 ATM. TO 0.1MBAR≤30MIN	
14	LEAKAGE RATE	0.01MBAR.L/S.	

UTILITY CONNECTION

CONN. #	SIZE	CONNECTION	DESCRIPTION
C1	1.5"	CLAMP	PURE STEAM INLET
C2	1.5"	CLAMP	WFI INLET
C3	3/4"	CLAMP	CITY WATER INLET FOR WATER RING PUMP
C4	DN50	FLANGE	MAIN DRAIN
C5	1"	CLAMP	JACKET COOLING CITY WATER INLET
C6	1"	CLAMP	JACKET COOLING CITY WATER OUTLET
C7	1"	CLAMP	CLEAN AIR INLET FOR AERATION SYSTEM
C8	1"	CLAMP	CLEAN N ₂ INLET FOR AERATION SYSTEM
C9	2"	CLAMP	COOLING WATER INLET FOR REFRIGERANT COOLING
C10	2"	CLAMP	COOLING WATER OUTLET FOR REFRIGERANT COOLING
C11	φ10	PUSH-IN FITTING	COMPRESSED AIR INLET FOR PNEUMATIC VALVE
C12	φ65	HOLE	POWER SUPPLY INLET
C13	1.5"	CLAMP	COOLING WATER INLET FOR VACUUM SYSTEM
C14	1.5"	CLAMP	COOLING WATER OUTLET FOR VACUUM SYSTEM
C15	φ51	TUBE	VACUUM EXHAUST
C16	φ10	PUSH-IN FITTING	DRY N ₂ INLET FOR VACUUM SYSTEM

EQUIPMENT SCHEDULE

EQUIP. #	QTY	FUNCTION	DESCRIPTION
E1	1	MAIN DOOR	TOFFLON
E2	1	SHELF HYDRAULIC CYLINDER	AS BUILT
E3	1	CHAMBER	TOFFLON
E4	1	MAIN ISOLATION VALVE	MUSHROOM VALVE
E5	1	CONDENSER	TOFFLON
E6	1	MUSHROOM VALVE CYLINDER	AS BUILT
E7	1	HYDRAULIC SYSTEM	ROKEN 40L
E8	1	VACUUM PUMP	EDWARDS GV80
E9	1	ROOTS PUMP	EDWARDS EH250
E10	1	CIRCULATION PUMP	WILO TOP-SD50/10
E11	1	ELECTRIC HEATER	12 KW
E12	2	PLATE HEAT EXCHANGER	ALFA LAVAL
E13	1	EXPANSION TANK	TOFFLON
E14	2	COMPRESSOR	BITZER S6J-16.2Y
E15	2	WATER CONDENSER	AS BUILT
E16	1	ELECTRICAL CABINET	AS BUILT
E17	1	LIQUID RING VACUUM PUMP	SPECK V95
E18	1	AERATION SYSTEM	TOFFLON
E19	1	PNEUMATIC UNITS	SMC
E20	1	PIZZA DOOR	TOFFLON
E21	1	UNLOADING PUSHER	TOFFLON

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TITLE: LYO-5 (SIP,CIP) GENERAL ARRANGEMENT			SHEET: 1 OF: 1		SIZE: A2	
			DWG: N/A		REV: 0.1	
CUSTOMER: CLIENT NAME			PROJECT NUMBER: TFL-GJTP-23-XXX			
DESIGNED: XXX	DATE: XXXX.XX.XX	MASS: 14,000 KG	TOLERANCED UNLESS OTHERWISE NOTED			
CHECKED: XXX	DATE: XXXX.XX.XX	SCALE: N/A		MACHINING	WELDING	SHEET METAL
APPROVED: XXX	DATE: XXXX.XX.XX	MATERIAL: N/A		GB/T 1804-m GB/T 1184-K	GB/T 1804-B GB/T 1804-F	GB/T 15055-m